MedRef v5.0 Vision

Architectural Blueprint & End-to-End Implementation Plan

Executive Summary

MedRef v5.0 transitions the MedRef platform from a single-corpus drug prescribing label system (**Layer 1 - Completed in Corpus v4.0 with 500 Therapeutics & 19,892 Vector Chunks**) into a multi-collection, patient-aware clinical decision intelligence system. By integrating disease guidelines (ADA, GOLD, GINA, ACC/AHA, KDIGO), specialized interaction databases, lab interpretation, clinical calculators, and conversational intake triage, MedRef v5.0 delivers 100% grounded clinical answers tailored to individual patient profiles.

1. The 4 Functional Layers of MedRef v5.0

Layer	Name	Function & Scope	Status
Layer 1	Drug Intelligence	Full RAG pipeline over 500 DailyMed SPL labels & openFDA metadata with 100% citation grounding.	**COMPLETE** (Corpus v4.0)
Layer 2	Disease Intelligence	Conversational symptom triage (duration, temp, age, red flags) providing differential assessments & safety disclaimers.	**PLANNED** (Phase 2)
Layer 3	Patient-Aware Engine	Context-aware pipeline incorporating Patient Profiles (age, eGFR, comorbidities, active meds) for renal dosing & interaction checks.	**PLANNED** (Phase 3)
Layer 4	Clinical Chat & Audit	ChatGPT-grade conversational UI with 10-point answer schema, multi-source badges ([DailyMed], [ADA]), & interactive evidence drawer.	**PLANNED** (Phase 4)

2. Multi-RAG Vector Collection Architecture

Instead of storing all clinical domain knowledge in a single vector collection, MedRef v5.0 establishes 6 specialized Qdrant vector collections managed by a high-precision **Intent Classifier Router (intent_router.py)**:

Collection Name	Target Knowledge Domain	Primary Sources
openfda_labels	500 Clinical Prescription Drug Labels	DailyMed SPL XML, openFDA, RxNorm
disease_corpus	Disease Practice Guidelines & Management	ADA 2026, GOLD, GINA, ACC/AHA, KDIGO, IDSA

drug_interactions	Drug-Drug & Drug-Disease Interaction Pairs	FDA Blackbox Warnings, DrugBank, MicroMedex
lab_values	Lab Interpretation & Reference Ranges	HbA1c, INR, Creatinine, Troponin, D-Dimer
clinical_calculators	Clinical Risk & Dosing Calculators	eGFR (CKD-EPI), CHA■DS■-VASc, HAS-BLED, Wells
patient_education	Plain-Language Patient Counseling	FDA Medication Guides, Mayo Clinic Education

3. Layer 3: Patient-Aware Execution Pipeline

When evaluating clinical scenarios for specific patients, the pipeline executes a multi-stage context-enrichment workflow:

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Patient Profile Context (Age: 67, Gender: M, CKD Stage 3, eGFR: 42, Meds: Warfarin, Losartan)
↓
Intent Router (Classifies query & routes to openfda_labels + drug_interactions + disease_corpus)
↓
Interaction Engine (Cross-checks active meds for drug-drug interactions, e.g. NSAID + Warfarin →
Bleeding Warning)
↓
Renal Dose Adjustment Checker (Evaluates drug against eGFR 42 mL/min using DailyMed
renal_impairment LOINC section)
↓
Multi-Collection Grounded Synthesizer (Generates 10-point answer schema with inline [DailyMed]
[KDIGO] citations)
↓
Trust & Audit Layer (Calculates groundedness score %, evidence diversity, & exposes raw vector
chunk drawer)
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4. Standardized 10-Point Answer Schema

To maintain absolute reliability and clinical consistency across all chat interactions, every generated answer adheres to a rigid 10-point schema:

1. **Overview**: High-level drug class / disease summary.
2. **Clinical Summary**: Grounded mechanism of action or pathophysiology.
3. **Symptoms / Indications**: FDA-approved indications or diagnostic criteria.
4. **Diagnosis / Testing**: Relevant lab interpretations or clinical calculators.
5. **Treatment Options**: First-line vs second-line therapies.
6. **Medications & Dosing**: Standard adult and renal/hepatic dose adjustments.
7. **Contraindications & Boxed Warnings**: Blackbox warnings and severe safety risks.
8. **Drug Interactions**: Co-administration risks and major drug-drug conflicts.

9. **Monitoring Parameters**: Required lab monitoring (e.g. INR, eGFR, LFTs).
10. **Patient Advice & References**: Plain-language counseling + inline multi-source citations.

5. Phased Implementation Roadmap

Phase	Key Deliverables	Verification Criteria
Phase 1 Multi-RAG & Router	• Build intent_router.py • Create disease_corpus & drug_interactions Qdrant collections • Index ADA, GOLD, KDIGO guidelines	• Intent classifier test suite passes with >95% accuracy • Multi-collection vector queries return relevant chunks
Phase 2 Disease Intelligence	• Implement intake_manager.py for conversational symptom triage • Build dialogue state tracker for symptom duration, temp, age, red flags	• Symptom triage flow correctly prompts for red flags & returns non-diagnostic disclaimers
Phase 3 Patient-Aware Engine	• Implement Patient Profile drawer & schema • Build clinical_evaluator.py for automated interaction & renal dose checks	• Patient with CKD Stage 3 + Warfarin asking for NSAID triggers acute renal & bleeding alerts
Phase 4 Clinical Chat UI	• Build ChatGPT-grade conversational React chat UI • Implement 10-point answer schema & multi-source citation badges ([DailyMed], [KDIGO])	• UI renders interactive evidence drawer, telemetry breakdown, & confidence badges